

① Nosocomial infections

These are called as hospital acquired infections (HAI).

The causative agents are bacteria which are opportunistic pathogens & most imp.

bacteria which causes nosocomial infections Staphylococcus aureus & Pseudomonas aeruginosa -

② Methanogens

① These are archaea

② They are autotrophs

③ They use CO_2 as final electron acceptor & as a sole carbon source.

④ They generate methane by reducing CO_2 .

⑤ They are strictly anaerobic bacteria -

③ Red Extreme Halophiles

These are halophilic bacteria

which require salt conc. (NaCl) 22-27% for their optimal ~~tem~~ growth. They are archaebacteria. They produce bacteriorhodopsin which gives them red colour.

Animalcules

These were 1st reported by Antonie van Leeuwenhoek.

When he observed some moving objects in the rain water & diff. body secretions with home made microscope. He named those moving objects as animalcules.

These were protozoa, diaphanous & spirochaetes.

Preservation by Overlaying culture

The bacterial culture on agar plates can be preserved by pouring some mineral oil

on the bacterial colonies. The addition of mineral oil retards the bacterial growth and helps bacteria to remain in same state. It is short term method for storage of culture.

Exponential Phase growth

- Log phase + Lag phase $\rightarrow$ definition
- Equation

$$\log_2 \left(\frac{N}{N_0} \right) = K_2 t$$

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Pressure

Barophilus $\rightarrow$ Loves

Bacteria which grow in pressurized environment.

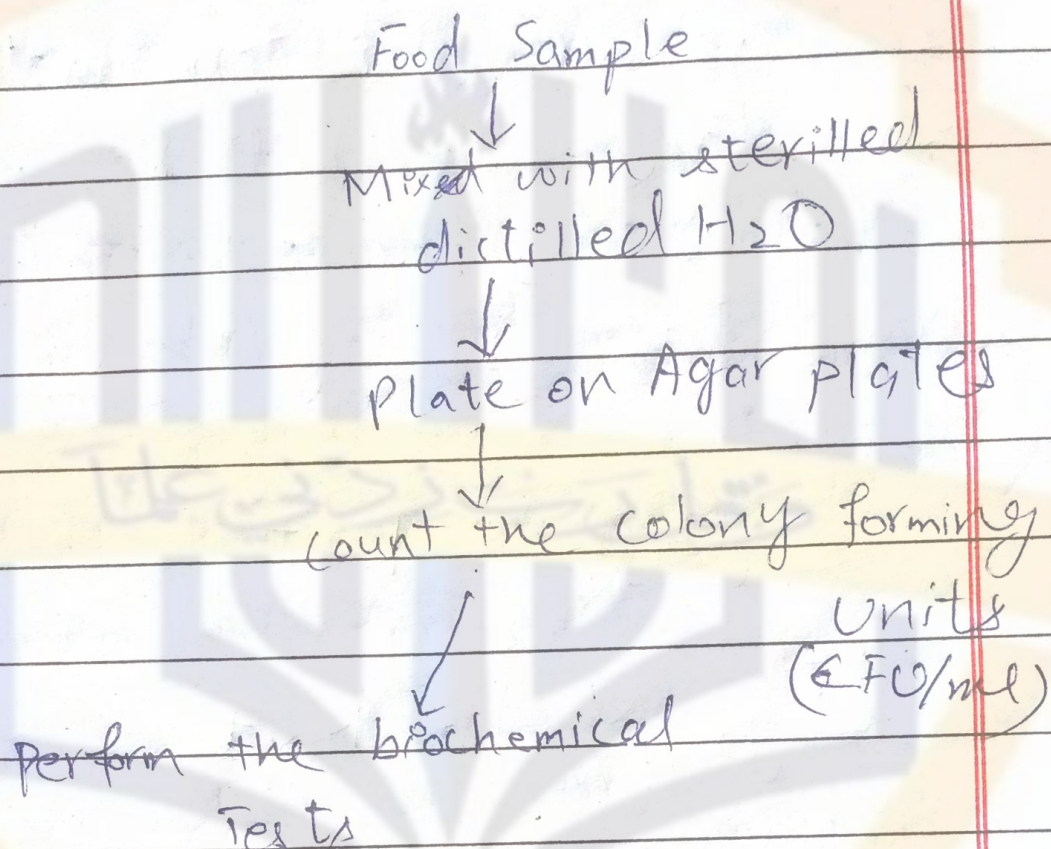
They are pressure loving bacteria & present deep in oceans.

They are archaeobacteria.

They are isolated from Mariana Trench.

e.g., *Shewanella*, *Methanococcus*, *Pyrococcus* & *Moritella* -

• General scheme of ^{bacteria} of food.



• Complementary System

Scheme for Coliform in water

from Book.

Bacterial sources of antimicrobial in Milk

- ① The milk can be contaminated comes from the disease called as mastitis which causes the

increase bacterial count in milk. plus pus & blood.

It is infection in the udder caused by *Staphylococcus aureus* which causes tissue damage in cows.

• **Activated Sludge** → waste water from book.

• Bactericidal	&	Bacteriostatic
↓		↓
a compound to kill bacteria		a compound to stop bacterial growth for some time
e.g., alcohol, phenol & iodine		e.g., tetracycline, penicillin, antiseptics

• **Phenol coefficient method:**

The method is used to check the lethal effects of phenol. It tells about bactericidal activity of phenol.

• By using this method we determine the lethality of diff. phenol dilutions. for a specific

time period.

• Minimum Lethal Dose (MLD)

The minimum conc. ^{or dose} of an agent which effectively kills the M.O.s called as M.L.D.

• Microbial Flora of Soil

① Protozoa ② Bacteriophages

③ Bacteria (Clostridium, Bacillus, E. coli, Pseudomonas, Klebsiella)

But it can be changed by human interferences.

• Attenuation:

It is done for the purpose of preparation of vaccine.

An attenuated $\leftrightarrow$ microbial culture is not able to cause infection but it can elicit

the immune response of the adoptive immunity & can cause production of antibodies & memory cells.

- It can be accomplished by removing certain structures from microbial cells.

• Secondary Infections

First the primary infection occurs and after its treatment the secondary infection can occur.

e.g., It can occur in case of skin infections, allergic reactions.

The Activated-Sludge Process. Vigorous aeration of sewage (Fig. 27-12) results in the formation of a floc: the finely suspended and colloidal matter of sewage forms aggregates designated as flocules. If this floc is permitted to settle and then added to fresh sewage that is again vigorously aerated, flocculation occurs in a shorter time than before. By repetition of this process, i.e., addition of sedimented floc to fresh sewage, aeration, sedimentation, addition of sediment

to fresh sewage, aeration, etc., a stage is reached where complete flocculation of the fresh sewage occurs very quickly, e.g., a few hours. These particles of floc, i.e., "activated sludge," contain large numbers of very actively metabolizing bacteria, together with yeasts, molds, and protozoa. This combination of microbial growth is very effective in the oxidation of organic compounds. A poor settlement of activated-sludge flocs adversely affects performance of a sewage-treatment plant. The sludge becomes more voluminous and is difficult to con-

The use of activated sludge is of great importance in wastewater treatment. This process usually employs an aeration period of 4 to 8 h, after which the mixture is piped to a sedimentation tank. The effluent from these tanks represents wastewater treated to secondary levels; there is a considerable reduction of suspended solids, and the BOD is reduced. Some of the advantages of this method of treatment are that relatively little land space is needed and the quality of final effluent is such that it does not require high dilution for disposal.

Activated sludge is a biological treatment process used in wastewater treatment plants to break down organic matter and remove pollutants from sewage or industrial wastewater. Here's an overview of activated sludge:

- 1. **Process**: In the activated sludge process, wastewater is mixed with a culture of microorganisms (referred to as activated sludge) in aeration tanks. These microorganisms are typically bacteria and some protozoa.*
- 2. **Aeration**: The aeration tanks are continuously aerated to provide oxygen to the microorganisms, facilitating their growth and metabolic activity. The oxygen supports aerobic degradation of organic pollutants present in the wastewater.*
- 3. **Biological Treatment**: The microorganisms in the activated sludge metabolize and consume organic pollutants as a food source. They break down complex organic compounds into simpler, more stable substances through biochemical reactions.*
- 4. **Floc Formation**: As microorganisms consume organic matter, they form aggregates or flocs, which settle more easily than individual cells. This facilitates the separation of the treated wastewater from the activated sludge in subsequent settling tanks.*
- 5. **Clarification**: After the aeration stage, the mixture of treated wastewater and activated sludge is directed to settling tanks, where the flocs settle to the bottom as sludge, while the clarified water is collected from the top.*
- 6. **Recycling and Disposal**: A portion of the settled sludge, known as return activated sludge, is recycled back to the aeration tanks to maintain an active population of microorganisms. Excess sludge is removed from the system and may undergo further treatment or disposal.*
- 7. **Effluent Quality**: The activated sludge process is effective in reducing organic pollutants, nutrients (such as nitrogen and phosphorus), and pathogens from wastewater, resulting in treated effluent that meets regulatory standards for discharge or reuse.*

Activated sludge is a widely used and effective method for wastewater treatment due to its ability to efficiently remove organic pollutants and improve water quality before discharge into the environment or further treatment for reuse purposes.